Assignment 7

CS 6117 (Fall 2025)

In this homework, we will build up a language and type system for metric spaces. Throughout, we will work with the category **Met** of *extended metric spaces*, i.e., sets X equipped with a distance $d : X \times X \to \mathbb{R}^+ \cup \{\infty\}$ satisfying:

- Reflexivity: $d(x, y) = 0$ if and only if $x = y$.
- Symmetry: $d(x, y) = d(y, x)$.
- Triangle inequality: $d(x, z) \leq d(x, y) + d(y, z)$.

Exercise 1. In A1 we saw two ways of defining a product on **Met**: we had $A \times B$ with the max distance, and $A \otimes B$ with the sum distance. We saw that $A \times B$ is the Cartesian product, and $A \otimes B$ is not. But $A \otimes B$ is a *monoidal* product.

1. Show that there is a functor $- \otimes - : \mathbf{Met} \times \mathbf{Met} \to \mathbf{Met}$, taking metric spaces (A, B) to $A \otimes B$. (This kind of functor is also known as a *bifunctor*.)
2. An object I is called a *unit* for the monoidal product $\otimes$ if we have natural isomorphisms $\lambda_X : I \otimes X \to X$ and $\rho_X : X \otimes I \to X$ (called *left* and *right* unitors). For a given monoidal product, the unit object is unique up to isomorphism. Describe the monoidal unit for the product $\otimes$.

A few notes:

- The functor $(- \otimes -)$ can also be equipped with natural transformations (unitors, associators, and symmetry) making $(\mathbf{Met}, 1, \otimes)$ into a *symmetric monoidal category* (SMC). You don't have to define and check these although they are all more or less immediate; you may want to take a look at nlab or wikipedia to see what these transformations should satisfy.
- When the monoidal unit is also the terminal object, the monoidal product is called *semicartesian*. These products are important (and common) because they allow weakening/discarding, giving an affine type system instead of a linear type system.

---

why not \otimes?

1. Given objects A, B in Met (with distances d_A, d_B), define $A \oplus B$ as: Underlying set: $A \times B = \{(a, b) \mid a \in A, b \in B\}$. Distance: $d_{A\oplus B}((a, b), (a', b')) = d_A(a, a') + d_B(b, b')$. Verify that $A \oplus B$ is indeed an object in Met: Reflexivity: $d_{A\oplus B}((a, b), (a', b')) = 0$ iff $d_A(a, a') + d_B(b, b') = 0$ iff (since distances are non-negative) $d_A(a, a') = 0$ and $d_B(b, b') = 0$ iff $a = a'$ and $b = b'$ iff $(a, b) = (a', b')$. Symmetry: $d_{A\oplus B}((a, b), (a', b')) = d_A(a, a') + d_B(b, b') = d_A(a', a)+d_B(b', b) = d_{A\oplus B}((a', b'), (a, b))$. Triangle inequality: For any $(a, b), (a', b'), (a'', b'') \in A \times B$,$d_{A\oplus B}((a, b), (a'', b'')) = d_A(a, a'') + d_B(b, b'') \leq [d_A(a, a')+d_A(a', a'')]+[d_B(b, b')+d_B(b', b'')] = d_{A\oplus B}((a, b), (a', b'))+ d_{A\oplus B}((a', b'), (a'', b''))$, using the triangle inequalities in A and B. Thus, $A \oplus B$ is a valid metric space.

   Given morphisms $f : A \to C$ and $g : B \to D$ in Met (both non-expansive), define $f \oplus g : A \oplus B \to C \oplus D$ as: On elements: $(f \oplus g)(a, b) = (f(a), g(b))$. Verify that $f \oplus g$ is a morphism in Met (i.e., non-expansive): $d_{C\oplus D}((f \oplus g)(a, b), (f \oplus g)(a', b')) = d_{C\oplus D}((f(a), g(b)), (f(a'), g(b'))) = d_C(f(a), f(a'))+d_D(g(b), g(b')) \leq d_A(a, a')+d_B(b, b') = d_{A\oplus B}((a, b), (a', b'))$, using the non-expansiveness of f and g.

   Preserves identities: For objects A, B, $\mathrm{id}_A \oplus \mathrm{id}_B : (a, b) \mapsto (a, b)$, which is exactly $\mathrm{id}_{A\oplus B}$ (and preserves the sum distance, as identities are non-expansive). Preserves composition: Given $f : A \to C$, $f' : C \to E$, $g : B \to D$, $g' : D \to F$, $(f' \circ f) \oplus (g' \circ g) : (a, b) \mapsto (f'(f(a)), g'(g(b)))$, and $(f' \oplus g') \circ (f \oplus g) : (a, b) \mapsto (f \oplus g)(a, b) = (f(a), g(b)) \mapsto (f'(f(a)), g'(g(b)))$, so they are equal as functions, and both preserve distances (by the previous steps).

   Thus, $- \oplus -$ is a functor from $\mathbf{Met} \times \mathbf{Met}$ to $\mathbf{Met}$.

2. The unit should satisfy $I \oplus X \cong X \cong X \oplus I$ via natural isomorphisms (natural in X). Consider the one-point metric space $I = 1 = \{*\}$ (a singleton set) with distance $d_1(*, *) = 0$. This is a valid metric space: reflexivity holds (only one point), symmetry trivial, triangle trivial.

   Underlying set of $I \oplus X$: $1 \times X = \{(*, x) \mid x \in X\}$, which is in bijection with X via $(*, x) \mapsto x$. Distance in $I \oplus X$: $d_{I\oplus X}((*, x), (*, x')) = d_1(*, *) + d_X(x, x') = 0 + d_X(x, x') = d_X(x, x')$. Thus, $I \oplus X$ is isometric (distance-preserving bijection) to X: Define $\lambda_X : I \oplus X \to X$ by $\lambda_X(*, x) = x$. This is bijective (inverse: $x \mapsto (*, x)$). Preserves distances: $d_X(\lambda_X(*, x), \lambda_X(*, x')) = d_X(x, x') = d_{I\oplus X}((*, x), (*, x'))$. Since isometries are isomorphisms in Met (non-expansive bijections with non-expansive inverses), λ_X is an iso. Similarly for $X \oplus I$: Distance: $d_{X\oplus I}((x, *), (x', *)) = d_X(x, x') + d_1(*, *) = d_X(x, x')$. Define $\rho_X : X \oplus I \to X$ by $\rho_X(x, *) = x$. Same reasoning: isometric iso.

   λ is natural in X: For any morphism $f : X \to Y$ in Met, Left side: $f \circ \lambda_X : (*, x) \mapsto f(x)$. Right side: $\lambda_Y \circ (\mathrm{id}_I \oplus f) : (*, x) \mapsto (*, f(x)) \mapsto f(x)$. They are equal. Similarly for ρ: $f \circ \rho_X = \rho_Y \circ (f \oplus \mathrm{id}_I)$.

   This is a standard result in monoidal categories: if I' is another unit with isos $\lambda'_X : I' \oplus X \to X$, $\rho'_X : X \oplus I' \to X$, then $I' \cong I$. The iso $I' \to I$

is $\rho'_I \circ \lambda'^{-1}_{I'} : I' \cong I' \oplus I \cong I$ (and similarly in reverse), and it commutes with the unitors (by naturality and monoidal axioms). In this case, any other unit I' must be isometric to the one-point space (since $I' \oplus X \cong X$ implies I' acts like a "neutral element" for the sum metric, forcing I' to contribute 0 to distances).

Exercise 2. Let $A \multimap B$ be the metric space with carrier the set of non-expansive maps and distance the supremum distance. On A3 you learned that this function space is not right-adjoint to the Cartesian product, so it was unclear if **Met** is Cartesian closed. However, it turns out to be *monoidal closed*. Show that we have the following isomorphism:

$$\mathbf{Met}(Z \otimes A, B) \cong \mathbf{Met}(Z, A \multimap B)$$

Thus, **Met** is a *symmetric monoidal closed category* (SMCC).

$A \multimap B$ has the supremum metric $d(f,g) = \sup_{a \in A} d_B(f(a), g(a))$.

Define currying $\Lambda : \mathbf{Met}(Z \oplus A, B) \to \mathbf{Met}(Z, A \multimap B)$ by $\Lambda(\phi)(z) = \lambda a.\phi(z,a)$. Verify $\Lambda(\phi)$ is non-expansive: for z, z', $d_{A \multimap B}(\Lambda(\phi)(z), \Lambda(\phi)(z')) = \sup_a d_B(\phi(z,a), \phi(z',a)) \leq d_Z(z,z')$, using ϕ's non-expansiveness.

Define uncurrying $\widetilde{\cdot} : \mathbf{Met}(Z, A \multimap B) \to \mathbf{Met}(Z \oplus A, B)$ by $\widetilde{\psi}(z,a) = \psi(z)(a)$. Verify $\widetilde{\psi}$ is non-expansive: $d_B(\widetilde{\psi}(z,a), \widetilde{\psi}(z',a')) \leq d_Z(z,z') + d_A(a,a')$, by triangle inequality and non-expansiveness of $\psi(z)$ and ψ.

They are inverses: $\Lambda(\widetilde{\psi}) = \psi$ and $\widetilde{\Lambda(\phi)} = \phi$. Naturality holds: for morphisms f, g, h, commute via pre/post-composition. Symmetry and associativity follow from commutative/associative addition in distances; unit is the singleton I. Coherence axioms are standard. Thus, **Met** is an SMCC, as $- \oplus A \dashv A \multimap -$.

Exercise 3. Given that a SMCC is close to a model of linear logic, it is a reasonable idea to see if we can define a comonad on the category **Met**. To help guide our search, we want to find a comonad $! : \mathbf{Met} \to \mathbf{Met}$ (plus associated natural transformations) such that $(!A) \multimap B$ corresponds to the set of *all* functions from A to B, without any restrictions. (Categorically, in models of linear logic the co-Kleisli category for the comonad is Cartesian closed, so intuitively these morphisms should behave like regular STLC.)

1. Define a suitable comonad: a functor $! : \mathbf{Met} \to \mathbf{Met}$, along with associated natural transformations. You do not need to spell out the diagrams.

   Your comonad should also satisfy the Seely isomorphism: $!(A \times B) \cong !A \otimes !B$ and there should be a natural transformation $m_{A,B} : !A \otimes !B \to !(A \otimes B)$ (in fact it should be symmetric monoidal). [Hint: it will be useful that ∞ is the maximum distance.]

2. Show that your comonad can be decomposed into $! = F \circ U$, where $U : \mathbf{Met} \to \mathbf{Set}$ and $F \dashv U$ are adjoints. Note that (1) **Set** is Cartesian, and (2) the decomposition ("resolution") of comonad into an adjunction is not unique.

---

1. Define $!X$ as the metric space with the same underlying set as X, but equipped with the metric $d_{!X}(x, y) = 0$ if $x = y$, and ∞ otherwise. This satisfies reflexivity (since $d = 0$ iff equal), symmetry, and the triangle inequality (as $\infty + r = \infty$ for any r, and $\infty \leq \infty$). On morphisms, $!f = f$ (underlying map). Non-expansiveness holds: $d_{!Y}(f(x), f(y)) \leq d_{!X}(x, y)$ is trivial for $x = y$. The counit $\varepsilon_X : !X \to X$ is the identity map, non-expansive as $d_X(x, y) \leq \infty$ for $x \neq y$. The comultiplication $\delta_X : !X \to !!X$ is the identity (since $!!X = !X$, as the metric is idempotent). These satisfy coassociativity ($\delta \circ \delta = !\delta \circ \delta$) and counit laws ($\varepsilon_! \circ \delta = id$, $!\varepsilon \circ \delta = id$), yielding a comonad.

   For the Seely isomorphism, $!(A \times B) \cong !A \oplus !B$, where $\times$ uses max distance: both sides have metric 0 if equal, ∞ otherwise (sum absorbs ∞, max does too for non-equal pairs). The transformation $m_{A,B} : !A \oplus !B \to !(A \oplus B)$ is the identity. This makes ! lax symmetric monoidal, with coherence from identities. Morphisms in the co-Kleisli category from A to B are $\mathbf{Met}(!A, B)$, bijective to all functions $A \to B$ (underlying), as the $0/\infty$ metric imposes no restrictions beyond reflexivity.

2. For decomposition, $! = F \circ U$, where $U : \mathbf{Met} \to \mathbf{Set}$ forgets the metric, and $F : \mathbf{Set} \to \mathbf{Met}$ equips sets with the $0/\infty$ metric. The adjunction $F \dashv U$ holds. $\mathbf{Set}(S, UX) \cong \mathbf{Met}(FS, X)$, as arbitrary functions $S \to X$ are non-expansive from FS. Then $! = F \circ U$, as $!X = F(UX)$. Set is Cartesian closed, and such decompositions are not unique

---

Exercise 4. It turns out that the comonad can be further refined into a family of comonads, called a *graded* comonad. For more on the structures described here, see "Combining Effects and Coeffects via Grading" by Gaboardi, Katsumata, Orchard, Breuvart, and Uustalu (ESOP 2016).

A *pre-ordered semiring* S consist of a set of elements, a preorder $\leq$, a commutative monoid $(+, 0)$, and a monoid $(\cdot, 1)$ such that both monoid operations are monotone with respect to the preorder (i.e., $a \leq a'$ and $b \leq b'$ implies $a + b \leq a' + b'$ and similarly for multiplication), and multiplication distributes over addition ($a \cdot (b + c) = a \cdot b + a \cdot c$). The example we will consider is the non-negative extended reals $\mathbb{R}^+ \cup \{\infty\}$ with the usual order and the usual addition and multiplication, operating on infinity as:

- $0 \cdot \infty = \infty \cdot 0 = 0$,

- $r \cdot \infty = \infty \cdot r = \infty$ for $r \neq 0$, and
- $\infty + r = r + \infty = \infty$.

We'll just call this semiring R for short. As usual, since R is a preorder, we can regard it as a posetal category $\mathbf{R}$ where objects are elements of R and there is a single morphism r to r' if and only if $r \leq r'$.

Define the following pieces of data, which form a *R-graded exponential comonad*:

1. Functors: $D : \mathbf{R} \to [\mathbf{Met}, \mathbf{Met}]$
2. 0-monoidality: $m_{r,I} : I \to Dr(I)$
3. 2-monoidality: $m_{r,A,B} : Dr(A) \otimes Dr(B) \to Dr(A \otimes B)$
4. Weakening: $w_A : D0(A) \to I$
5. Contraction: $c_{r,s,A} : D(r + s)(A) \to Dr(A) \otimes Ds(A)$
6. Dereliction: $\epsilon_A : D1(A) \to A$
7. Digging: $\delta_{r,s,A} : D(r \cdot s)(A) \to Dr(Ds(A))$

We'll briefly sketch the first map: the functor should map the object $r \in \mathbf{R}$ (i.e., a non-negative real number) to the endofunctor on **Met** that *scales* the metric by a factor of r and leaves morphisms unchanged; check that this is a functor. The rest of the maps are almost all trivial.

The subscripts suggest that these maps actually form components of natural transformations, but we will not check naturality (because all of them are obvious). Similarly, these operations should make a bunch of diagrams commute, but we will not check them here (because they are all straightforward).

---

1. D_r with transformations $D_{r'} \to D_r$ when $r \leq r'$, via identity where possible, non-expansive as $r \cdot d \leq r' \cdot d$.

   For $r > 0$ finite, $D_r(A)$ has underlying set of A, metric $d_{D_r(A)}(x, y) = r \cdot d_A(x, y)$. For $r = 0$, $D_0(A) = I$ (singleton, constant functor). For $r = \infty$, $D_\infty(A)$ has underlying set of A, metric ∞ if $x \neq y$, 0 otherwise. On morphisms, $D_r(f) = f$ for $r > 0$, constant for $r = 0$.
2. Identity on the unit I (singleton), as $r \cdot 0 = 0$.
3. Identity, since $d_{D_r(A) \oplus D_r(B)}((a, b), (a', b')) = r \cdot d_A(a, a') + r \cdot d_B(b, b') = d_{D_r(A \oplus B)}((a, b), (a', b'))$.
4. Unique map to terminal I, non-expansive as $D_0(A)$ is indiscrete (all distances 0).
5. Identity, as metrics match: $(r + s) \cdot d_A = r \cdot d_A + s \cdot d_A$.
6. Identity, since $D_1(A) = A$.
7. Identity, as $D_r(D_s(A))$ scales by $r \cdot s$.

Exercise 5. Now that we have identified lots of interesting structure in **Met**, we can build a language and type system where well-typed programs are interpreted as morphisms in this category, putting these structures to work. The language we will sketch is a variant of the Fuzz language proposed by Reed and Pierce (ICFP 2010), and subsequently extended by other researchers.

Given base types α, types in Fuzz are generated by the following grammar:

$$\tau ::= \alpha \mid () \mid \tau \otimes \tau \mid \tau \multimap \tau \mid !_r\tau \ (r \in R)$$

If we interpret each base type α as a metric space in **Met**, then we can interpret every type as an object in **Met** using the categorical constructions from the previous exercises. Typing contexts have the form

$$\Gamma = x_1 :_{r_1} \tau_1, \ldots, x_n :_{r_n} \tau_n$$

where each $r \in R$ is called a *sensitivity* annotation.

Notably, typing contexts can be added: given typing contexts Γ and Δ assigning the same types to variables, we can define $\Gamma + \Delta$ by adding the sensitivities (e.g., $(x :_r \tau) + (x :_s \tau) = x :_{r+s} \tau$). Typing contexts can also be scaled: given a context Γ and a scaling factor $r \in R$, the context $r \cdot \Gamma$ scales each sensitivity in Γ by r (e.g., $r \cdot (x :_s \tau) = x :_{r \cdot s} \tau$).

The typing judgment $\Gamma \vdash e : \tau$ states that e has type τ in context Γ. We wish to interpret each well-typed term as a morphism in **Met**:

$$[\![\Gamma \vdash e : \tau]\!] : Dr_1[\![\tau_1]\!] \otimes \cdots \otimes Dr_n[\![\tau_n]\!] \to [\![\tau]\!].$$

where Dr is the scaling functor from the previous exercise. The empty context Γ is interpreted as the monoidal unit I. This way, each well-typed term has *sensitivity* described by the annotations r: if we vary the inputs of a program by distance δ, then the output should vary by distance at most $r \cdot \delta$.

Using any of the maps from the previous exercises, define the semantics of the following two rules:

1. Tensor introduction:

$$\frac{\Gamma \vdash e_1 : \tau_1 \qquad \Delta \vdash e_2 : \tau_2}{\Gamma + \Delta \vdash e_1 \otimes e_2 : \tau_1 \otimes \tau_2}$$

2. Tensor elimination

$$\frac{\Gamma \vdash e : \tau_1 \otimes \tau_2 \qquad \Delta, x :_r \tau_1, y :_r \tau_2 \vdash f : \tau}{r \cdot \Gamma + \Delta \vdash \text{let } (x \otimes y) = e \text{ in } f : \tau}$$

That is, assuming that we are given the interpretation of the premises, define a morphism interpreting the conclusion. To keep things readable, you can assume that Γ and Δ are compatible (have the same variables and types, but possibly different sensitivities). You should also omit any maps for reassociating or permuting the tensor product when managing contexts.

The denotational semantics interprets types as objects in **Met** and terms as non-expansive morphisms, with sensitivities r corresponding to Lipschitz constants (distance output $\leq r\cdot$ distance input). Base type $[\alpha]$ is a fixed metric space. $[I]$ is the singleton space. $[\tau \otimes \sigma] = [\tau] \oplus [\sigma]$ (sum metric). $[!r\tau] = D_r([\tau])$ (scaled metric by r). Context $[\Gamma] = [\tau_1] \oplus \cdots \oplus [\tau_n]$ (for $\Gamma = z_1 : \tau_1, \ldots, z_n : \tau_n$). Term $\Gamma \vdash e : r\tau$ is a morphism $[e] : [\Gamma] \to [\tau]$ with Lipschitz $\leq r$.

1. Tensor Introduction:

not the same objects: need contraction

$$[\Gamma + \Delta \vdash \langle e_1, e_2 \rangle : r\tau_1 \otimes \tau_2] = [e_1] \oplus [e_2] : [\Gamma] \oplus [\Delta] \to [\tau_1] \oplus [\tau_2]$$

2. Tensor Elimination:

$$[\Gamma + \Delta \vdash \text{let } \langle x, y \rangle = e \text{ in } f : r\rho] = [f] \circ (\text{id}_{[\Delta]} \oplus [e]) \circ \gamma_{[\Gamma],[\Delta]} : [\Gamma] \oplus [\Delta] \to [\rho]$$

where γ is the symmetry isomorphism

there should be a scaling here I think

Exercise 5 (Sparkly Star). Prove Proposition 1 in Benton's *A Mixed Linear and Non-Linear Logic: Proofs, Terms and Models*: if $F \dashv G$ is a monoidal adjunction of monoidal functors, then the left-adjoint F must be a strong monoidal functor. Benton's paper spells out the isomorphism; you just have to check the details. This is part of a broader class of general results called *doctrinal adjunctions* from higher category theory, which study adjunctions as pairs of morphisms making certain diagrams in a *2-category* (like **Cat**) commute.

I downgraded this question to an optional exercise since the full proof is in the full version of the LNL paper on EdStem

Not sure but try after reading ideas from nLab: Consider monoidal categories $\mathcal{C}$ and $\mathcal{D}$, with $F : \mathcal{C} \to \mathcal{D}$ left adjoint to $G : \mathcal{D} \to \mathcal{C}$, both lax monoidal, and the adjunction monoidal. By the doctrinal adjunction theorem, the lax monoidal structure on the right adjoint G induces a unique oplax monoidal structure on the left adjoint F such that η and ϵ are monoidal transformations. The oplax structure on F is given by the mate of the lax structure on G. Specifically, the lax structure on G consists of natural transformations: $n_0 : G(I_{\mathcal{D}}) \to I_{\mathcal{C}}$, and $n_{X,Y} : G(X \otimes_{\mathcal{D}} Y) \to GX \otimes_{\mathcal{C}} GY$. The mate induces the oplax structure on F: $\tilde{m}_0 : I_{\mathcal{D}} \to F(I_{\mathcal{C}})$, and $\tilde{m}_{A,B} : F(A \otimes_{\mathcal{C}} B) \to FA \otimes_{\mathcal{D}} FB$, via the adjunction's unit and counit (explicitly, $\tilde{m}_{A,B}$ is the transpose of $n_{FA,FB} \circ G(\epsilon_A \otimes_{\mathcal{D}} \epsilon_B) \circ \eta_{G(FA \otimes_{\mathcal{D}} FB)}$.

Since the adjunction is monoidal and F is also given as lax monoidal (with structure $m_0 : I_{\mathcal{D}} \to F(I_{\mathcal{C}})$, $m_{A,B} : FA \otimes_{\mathcal{D}} FB \to F(A \otimes_{\mathcal{C}} B)$), the doctrinal theorem implies that the lax structure m on F must be the inverse of the induced oplax structure $\tilde{m}$. Thus, m is an isomorphism, making F strong monoidal. This holds by the propositions in the nLab entry on doctrinal adjunction: for the unit and counit to be monoidal transformations, the oplax structure on F must have an inverse that coincides with its lax structure, hence F is strong. The mate ensures uniqueness and compatibility.

Exercise 6 (Category of the Week). Let X and Y be sets. We'll say that a *lens* from X to Y consists of (1) a *forward* or "get" map $f : X \to Y$ and a (2) *backward* or "update" or "put" map $b : X \times Y \to X$ such that the following lens laws hold:

- PutGet: $f(b(x, y)) = y$
- GetPut: $b(x, f(x)) = x$
- PutPut: $b(b(x, y), y') = b(x, y')$

There are many flavors of lenses, with different kinds of forward/backward maps, and different collections of lens laws. But all lenses have some kind of forward flow of information, paired with a corresponding backward flow of information. Categories of lenses often have useful structure.

1. Show that we can define a category **Lens** where objects are sets, and a morphism from X to Y is a lens from X to Y. Make sure to check that your definition of composition is well-defined (produces a valid morphism) and check the category laws carefully.
2. Show that we can define a bifunctor $(- \otimes -) : \mathbf{Lens} \times \mathbf{Lens} \to \mathbf{Lens}$ sending the pair of sets (A, B) to the set $A \times B$. This product makes **Lens** into a symmetric monoidal category.
3. Show how to define a lens from $A \otimes B \to A$.
4. Try to define a diagonal lens $A \to A \otimes A$. What seems to go wrong?

In general, lens categories are symmetric monoidal categories, but often don't have Cartesian products; thus, lens categories tend to support some kind of linear type system. Internal homs for lens categories tend to exist only in very special situations. There are even models of full linear logic in lens categories (see Valeria de Paiva's thesis).

---

1. Objects are sets. A morphism (lens) from set X to set Y is a pair $(f : X \to Y, b : X \times Y \to X)$ satisfying: PutGet: $f(b(x, y)) = y$ for all $x \in X$, $y \in Y$. GetPut: $b(x, f(x)) = x$ for all $x \in X$. PutPut: $b(b(x, y), y') = b(x, y')$ for all $x \in X$, $y, y' \in Y$.

   For $(f_1, b_1) : X \to Y$ and $(f_2, b_2) : Y \to Z$, the composite is get $f = f_2 \circ f_1$, put $b(x, z) = b_1(x, b_2(f_1(x), z))$. PutGet: $f(b(x, z)) = f_2(f_1(b_1(x, b_2(f_1(x), z)))) = f_2(b_2(f_1(x), z)) = z$. GetPut: $b(x, f(x)) = b_1(x, b_2(f_1(x), f_2(f_1(x)))) = b_1(x, f_1(x)) = x$. PutPut: $b(b(x, z), z') = b_1(b(x, z), b_2(f_1(b(x, z)), z')) = b_1(b_1(x, b_2(f_1(x), z)), b_2(f_1(b_1(x, b_2(f_1(x), z))), z')) = b_1(x, b_2(f_1(x), z')) = b(x, z')$.

   Identity on X: get id_X, put $b(x, x') = x'$. PutGet: $\mathrm{id}_X(x') = x'$. GetPut: $b(x, \mathrm{id}_X(x)) = x$. PutPut: $b(x', x'') = x'' = b(x, x'')$.

Associativity: Puts: For three lenses, the nested b definitions associate by function composition. Composition with identity leaves lenses unchanged. Thus, Lens is a category.

2. On objects: $A \otimes B = A \times B$. On morphisms: For $(f_1, b_1) : X \to A$, $(f_2, b_2) : Y \to B$, $(f_1, b_1) \otimes (f_2, b_2) : X \times Y \to A \times B$ has get $(x, y) \mapsto (f_1 x, f_2 y)$, put $((x, y), (a, b)) \mapsto (b_1(x, a), b_2(y, b))$. Laws hold componentwise. Symmetric monoidal: Unit $I = \{*\}$; associator $\alpha : ((a, b), c) \mapsto (a, (b, c))$, lens get/put id (bijection, laws hold); unitors $\lambda : (*, a) \mapsto a$, ρ similar; braiding $(a, b) \mapsto (b, a)$. Coherence from **Set**'s $\times$.

   usually called symmetry since it is self inverse

3. Get: $(a, b) \mapsto a$; put: $((a, b), a') \mapsto (a', b)$. PutGet $a' = a'$; GetPut $(a', b) = (a', b)$ wait no: GetPut $((a, b), a) = (a, b)$; PutPut $((a', b), a'') = (a'', b) = ((a, b), a'')$. Non-expansive implicitly (sets). This is a projection.

4. Assume $(f, b) : A \to A \times A$. For diagonal, get $f(a) = (a, a)$. PutGet: $f(b(a, (a_1, a_2))) = (a_1, a_2)$, so $b(a, (a_1, a_2)) = a'$ with $f(a') = (a_1, a_2) = (a', a')$, implying $a_1 = a_2$. Fails for $a_1 \neq a_2$. GetPut: $b(a, (a, a)) = a$; PutPut consistent only if components tied, but arbitrary pairs break it. No such lens exists.
